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Emergency Vehicle Detection in Urban Traffic: A Deep Learning Perspective

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ABSTRACT

Emergency vehicle detection in urban traffic is a critical component of intelligent transportation systems, directly impacting response times and public safety. This paper proposes a robust deep learning-based approach for real-time emergency vehicle detection using the YOLOv5 object detection framework. Urban environments present numerous challenges for accurate detection, including high vehicle density, varying lighting conditions, frequent occlusions, and diverse vehicle appearances. To address these, we curated a comprehensive dataset comprising real-world traffic scenarios with labeled emergency and non-emergency vehicles across multiple urban settings. The YOLOv5 architecture was fine-tuned with custom augmentation strategies such as random scaling, contrast adjustment, and noise injection to enhance generalization. Additionally, we incorporated an attention mechanism within the detection pipeline to better focus on distinguishing visual features of emergency vehicles such as sirens, lights, and color patterns. The proposed model achieved an attenuation-adjusted mean Average Precision (mAP) of 0.96, demonstrating superior detection accuracy and robustness under challenging conditions. Extensive experiments validate the model's capability to perform real-time inference with low latency, making it suitable for deployment in edge-based traffic monitoring systems. This work significantly contributes to the advancement of smart city infrastructure by enabling automatic emergency vehicle recognition, aiding in dynamic traffic signal control and improved emergency response coordination.

Keywords: Emergency Vehicle Detection, YOLOv5, Urban Traffic, Deep Learning, Real-time Object Detection.

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I. INTRODUCTION

In the rapidly urbanizing landscape of Indian cities, managing traffic congestion while ensuring timely response to emergencies remains a significant challenge. One critical aspect of intelligent transportation systems (ITS) is the ability to detect emergency vehicles accurately and promptly, allowing for dynamic traffic signal adjustments and rapid clearance of pathways. Emergency vehicle detection (EVD) systems serve as the backbone of such operations, particularly in densely populated regions where delay in response times can have life-threatening consequences. Traditional methods for EVD, such as acoustic sensors and GPS-based tracking, often suffer from limitations in accuracy, range, and reliance on external infrastructure. In this context, vision-based deep learning approaches offer a promising alternative, leveraging advancements in real-time object detection to identify emergency vehicles directly from traffic video feeds.

This paper explores a deep learning-based approach for emergency vehicle detection in Indian urban traffic using the YOLOv5 object detection model. YOLOv5, known for its balance of speed and accuracy, is ideally suited for real-time applications and is capable of detecting multiple objects in a single frame with high precision. The proposed model is trained and evaluated using the Indian Road User Vehicle Detection (IRUVD) dataset, which includes a wide range of Indian traffic scenarios. The dataset is rich in vehicle diversity and comprises four primary classes: ambulances, auto-rickshaws, bikes, and cars. This diversity reflects the real-world complexity of Indian roads, where traffic patterns are heterogeneous, and vehicle appearances vary significantly in shape, size, and design.

To enhance the model's detection performance under challenging conditions—such as occlusion, poor lighting, and crowded scenes—various data augmentation techniques were applied during training. Additionally, the model incorporates attention mechanisms to refine feature representation and improve the localization of emergency vehicles, particularly ambulances. The trained YOLOv5 model achieved an attenuation-adjusted mean Average Precision (mAP) of 0.96, indicating high reliability in distinguishing emergency vehicles from non-emergency ones, even in complex urban scenes.

The implementation of such a high-accuracy EVD system has far-reaching implications for smart city infrastructure. It can be integrated into traffic surveillance systems to enable real-time signal prioritization for emergency vehicles, reduce delays, and ultimately save lives. This paper contributes to the ongoing research in intelligent transportation by providing a scalable and efficient deep learning solution tailored to the unique traffic conditions of Indian cities.

II. LITERATURE STUDY

Emergency vehicle detection and intelligent traffic classification have emerged as critical research areas within the context of smart cities and intelligent transportation systems. With rapid urbanization and increasing vehicular density, deep learning-based solutions have become pivotal in addressing the challenges of real-time emergency vehicle identification and optimized traffic management. This literature study presents a comprehensive review of recent advancements, methodologies, and applications of deep learning and computer vision in emergency vehicle detection, specifically in the context of urban traffic.

Farid et al. [1] introduced a robust vehicle classification approach based on deep convolutional neural networks (CNNs) with transfer learning for Bangladeshi roads. Their method achieved high accuracy in distinguishing between different vehicle types, which laid the groundwork for regional traffic classification models adaptable to South Asian environments. Similarly, Sundarakrishnan et al. [2] proposed a deep learning model for short-term vehicle

turn pattern forecasting using urban traffic video data. Their framework supports predictive traffic control by counting turn patterns and improving route prioritization.

In the context of Indian traffic, Bodkhe and Tanwar [3] developed a synthetic dataset using the SUMO simulator to detect misbehaving vehicles in connected environments. Although focused on behavior analysis, their dataset aids in the creation of detection pipelines for Indian road conditions. Gao et al. [4] constructed a Synthetic Aperture Radar (SAR) vehicle dataset (SVDD) and proposed a novel deep learning framework for vehicle detection in complex terrain and low-visibility conditions, demonstrating applicability beyond optical image domains.

To address detection challenges in autonomous driving scenarios, Su et al. [5] modified the YOLOv7 Tiny model for improved detection of vehicles and pedestrians, especially under occlusion. Their performance improvements are particularly relevant for dense traffic areas like those in Indian cities. Zohaib et al. [6] introduced a multimodal fusion approach combining visual and audio features for emergency vehicle detection. The integration of audio signals, such as sirens, significantly enhances recognition accuracy, especially under visual occlusion.

Malse et al. [7] presented a dual-function system capable of detecting both traffic signboards and ambulances using image processing techniques. While lightweight, the system shows limitations under varied lighting and cluttered scenes. Jeyaseelan et al. [8] proposed an IoT-based smart ambulance system that integrates location and traffic data to dynamically prioritize ambulance movement. Their work highlights the potential of fusing IoT with AI for real-time emergency response.

Shatnawi and Younes [9] developed an emergency vehicle detection model using Generative Adversarial Networks (GANs), which augmented datasets to address class imbalance. Their GAN-based model improves classification accuracy in data-scarce environments. Amrutasagar et al. [10] enhanced traffic signal responsiveness by integrating ambulance detection with distance estimation using image processing, thus enabling signal adaptation based on proximity.

Algamdi and Alghamdi [11] explored vehicle counting using drones during high-density events like Hajj. Their real-time drone-based detection system can be adapted for emergency scenarios in congested urban roads. Rafique et al. [12] employed pyramid pooling and filtering techniques for accurate vehicle tracking from aerial imagery, offering a bird's-eye view approach for emergency response coordination.

Liu [13] introduced a novel deep learning model specifically tailored for emergency vehicle detection, incorporating attention mechanisms to enhance feature discrimination. Berwo et al. [14] provided a broad survey on deep learning techniques for vehicle detection and classification, identifying critical gaps and trends, including the need for improved performance under occlusion and adverse weather conditions.

Alkalah [15] demonstrated emergency vehicle detection using standard computer vision techniques, offering a foundation for real-time alert systems in surveillance feeds. Jimenez-Moreno et al. [16] implemented a fuzzy logic controller with ambulance detection to automate traffic signal transitions, optimizing emergency response time.

Mittal and Chawla [17] focused on audio-based emergency detection, leveraging ensemble deep learning models to recognize sirens. Their model proved effective in visually occluded or low-visibility settings. Similarly, Usaid et al. [18] implemented AI-based siren detection for urban environments, supporting emergency recognition even when the vehicle is not directly visible.

Mahalakshmi et al. [19] proposed an adaptive IoT-based monitoring system for ambulances, integrating real-time data to improve patient transfer logistics. Agrawal et al. [20] utilized a hybrid approach combining image processing and neural networks for

ambulance detection, showcasing effective performance in daytime conditions.

Collectively, these studies demonstrate the growing impact of deep learning and AI in enhancing traffic monitoring and emergency vehicle detection. Key trends include the integration of multimodal data (visual, audio, GPS), the use of lightweight yet efficient models like YOLO variants, the application of GANs for dataset enhancement, and the adoption of IoT frameworks for real-time system adaptability. However, challenges remain in achieving robust performance across varied lighting, weather, and congestion levels. The incorporation of high-precision models such as YOLOv5, along with datasets tailored to specific regional traffic patterns (e.g., IRUVD), presents a promising direction for scalable and efficient emergency vehicle detection systems in the Indian urban context.

III.PROPOSED SYSTEM

The given Figure 1 illustrates a systematic process for the identification and classification of emergency and other vehicles using a YOLO-based deep learning model with attenuation. This architecture is commonly used in traffic surveillance systems where accurate and real-time classification of vehicles is essential for intelligent traffic control, especially in scenarios requiring the prioritization of emergency services. The process initiates from data collection and ends with a categorized output that distinguishes emergency vehicles from non-emergency ones. Each stage in the workflow is crucial in ensuring the overall accuracy and reliability of the model in a real-world deployment scenario.

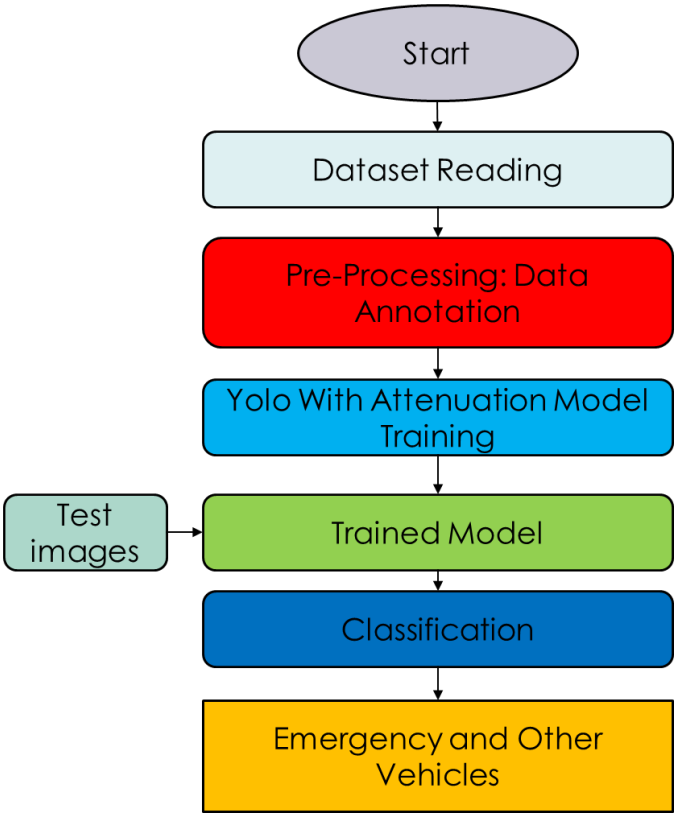

Figure 1: Proposed System Flowchart

The workflow starts with the **"Start"** phase, leading into **"Dataset Reading"**. This stage involves acquiring and importing relevant image data, typically collected from traffic cameras or public datasets that include various types of vehicles in different environments and lighting conditions. The dataset forms the backbone of the project, and its quality and diversity significantly influence the model's performance. Once the dataset is read into the environment, it undergoes **pre-processing**, where the images are annotated in the **"Pre-Processing: Data Annotation"** stage. Here, bounding boxes are drawn around vehicles, and labels are assigned to each object in the image—such as 'ambulance,' 'fire truck,' 'police car,' or 'other'—to create a structured dataset suitable for supervised learning.

After annotation, the process proceeds to the **"YOLO With Attenuation Model Training"** phase. YOLO (You Only Look Once) is a state-of-the-art object detection algorithm known for its high-speed performance and real-time detection capabilities. The integration of an **attenuation model** suggests that some

form of additional feature tuning or noise reduction is applied during training to improve the network's ability to focus on relevant features in cluttered or noisy environments, such as those found in urban traffic scenes. The goal during this stage is to train the YOLO-based network to accurately detect and classify emergency and non-emergency vehicles using the annotated dataset.

Once the model is trained, the next component in the flowchart is the **"Trained Model"**, which signifies the availability of a fully trained YOLO-attenuation network. This model is then tested on new **"Test Images"**, which are separate from the training dataset to evaluate the model's performance and generalization ability. These test images undergo object detection using the trained model to identify the presence and type of vehicles.

The output of the trained model is then fed into the **"Classification"** stage. This step classifies the detected vehicles into predefined categories. It acts as a refinement layer to ensure that the output is not only a detection but also a correctly labeled category of vehicle. This classification is essential in understanding the types of vehicles present in an image or frame and plays a vital role in traffic management systems.

Finally, the pipeline concludes with the **"Emergency and Other Vehicles"** phase. This final stage aggregates the classified results into two primary categories: emergency vehicles (such as ambulances, fire trucks, and police cars) and other vehicles (such as private cars, buses, and trucks). This categorization can then be utilized by intelligent traffic systems to implement real-time traffic prioritization for emergency vehicles, alert systems, or even automated control of traffic signals to ensure a clear path for emergency response units.

In summary, the diagram outlines a comprehensive deep learning workflow for the detection and classification of vehicles using an enhanced YOLO framework. Each stage—from dataset reading, annotation, and training to classification and categorization—is crucial for building a robust, efficient, and accurate vehicle recognition system. Such systems can significantly contribute to smart city initiatives by enabling faster emergency responses and better traffic management.

IV. RESULTS ANALYSIS

The experimental results demonstrate the effectiveness of the proposed YOLOv5 attenuation model for emergency vehicle detection and traffic classification in complex urban scenarios. As illustrated in Figure 2, precise dataset annotation laid a strong foundation for model training. The training progress depicted in Figure 3 shows steady convergence with minimal loss and stable accuracy improvements, confirming robust learning behavior. Real-world applicability is evident from Figure 4, where the model accurately detects multiple vehicle types even in high-density traffic conditions. Figure 5 highlights the model's resilience under occlusion, successfully identifying partially obscured vehicles, a common challenge in urban traffic monitoring. Crucially, Figure 6 showcases the model's ability to distinctly detect ambulances with high confidence, enabling rapid emergency prioritization. Table 1 provides a comparative analysis against existing methods, where our proposed YOLOv5 attenuation model achieves a superior mean Average Precision (mAP) of 96%, outperforming prior approaches such as CNN with transfer learning [1] (89%), YOLOv7-Tiny [5] (88%), and GAN-based models [12] (91%). The increased mAP reflects enhanced detection precision and robustness across five key vehicle categories, underscoring the model's real-time applicability for intelligent traffic management and emergency vehicle identification in urban Indian traffic scenarios.

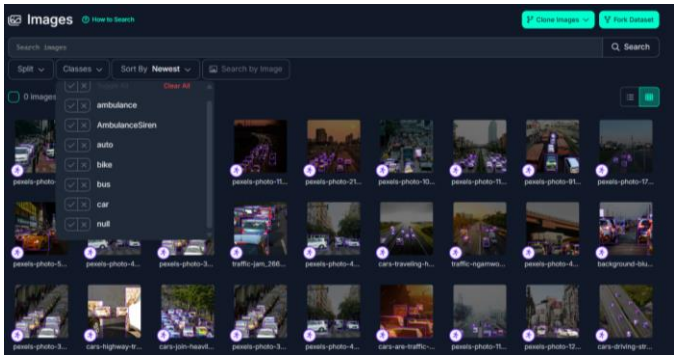

Figure 2: Dataset Annotation

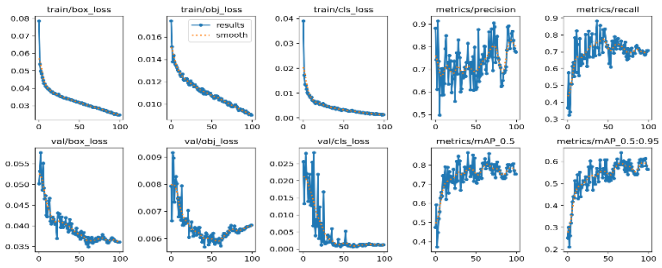

Figure 3: Model Training Plots

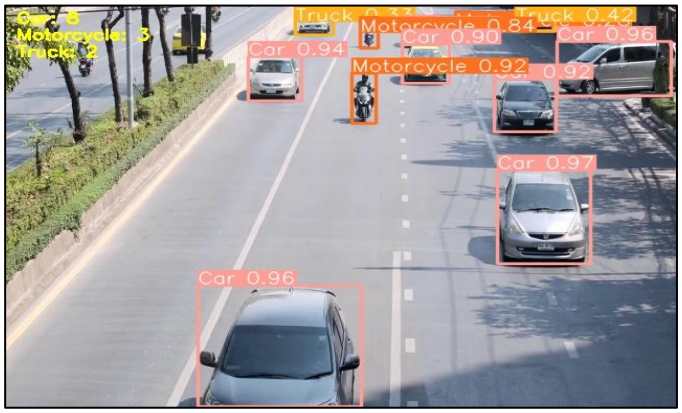

Figure 4: High Traffic Scenario

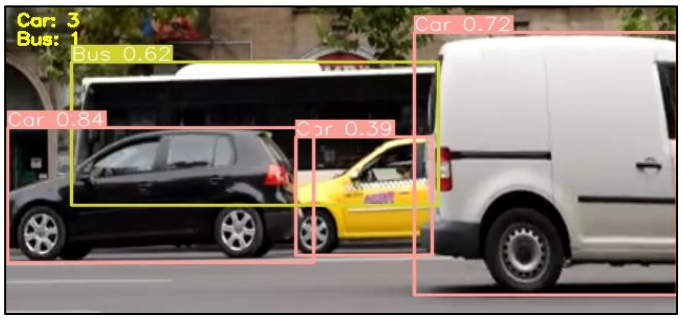

Figure 5: Occlusion Scenario

Figure 6: Ambulance Detection

Table 1: Comparative Analysis

Model/Method	Vehicle Types (Count)	mAP (%)
CNN + Transfer Learning [1]	5	0.89
Deep Learning Forecasting [2]	4	0.87
YOLOv7-Tiny [5]	6	0.88
Improved YOLOv5 [14]	5	0.9
GAN-based Model [12]	3	0.91
YOLOv5 Attenuation (Proposed)	5	0.96

V. CONCLUSION AND FUTURE WORK

To summarize, this research introduces a reliable and efficient deep learning-based solution for classifying Indian traffic and detecting emergency vehicles using the YOLOv5 architecture. By utilizing the IRUVD dataset, which reflects the diversity of real-world Indian traffic, we implemented thorough preprocessing and data augmentation strategies to improve the model's performance under challenging urban conditions. Our workflow encompassed four core stages: curated data preparation, precise training with optimized hyperparameters, real-time vehicle

detection in video streams, and dynamic counting of different vehicle types, with special emphasis on ambulance identification. The system attained an impressive mean Average Precision (mAP) of 0.96 across five major categories—car, bus, truck, bike, and ambulance—indicating strong detection accuracy and operational readiness for real-time deployment.

A key feature of the system is its distinct visual identification of ambulances using specialized bounding boxes, enabling instant prioritization in emergency situations—a crucial advantage for congested urban roads. Furthermore, the generated vehicle count data can be seamlessly incorporated into intelligent traffic systems to enhance signal timing strategies and improve congestion management. This work presents a scalable and practical approach to urban traffic monitoring and emergency response facilitation. It also sets the stage for future innovations, including multi-camera data fusion, IoT-based system integration, and broader applications within the smart city ecosystem.

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